

Accurate identification and mechanistic evaluation of pathogenic missense variants with Rhapsody-2

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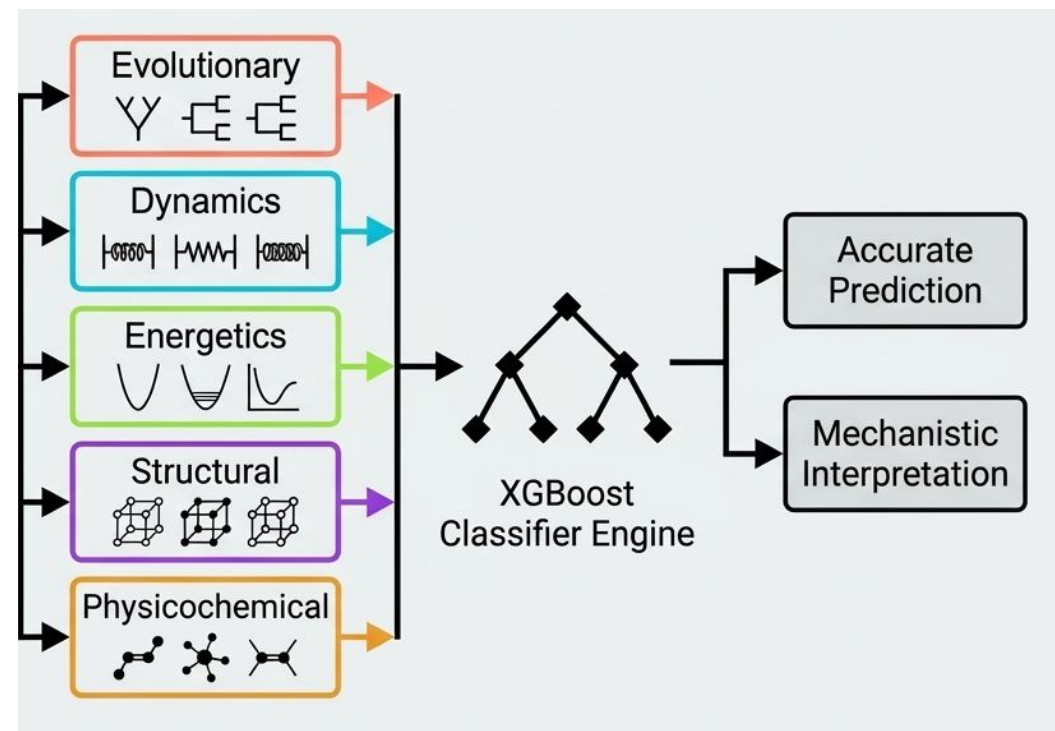
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Introduction and Motivation

Single amino acid variants (SAVs)

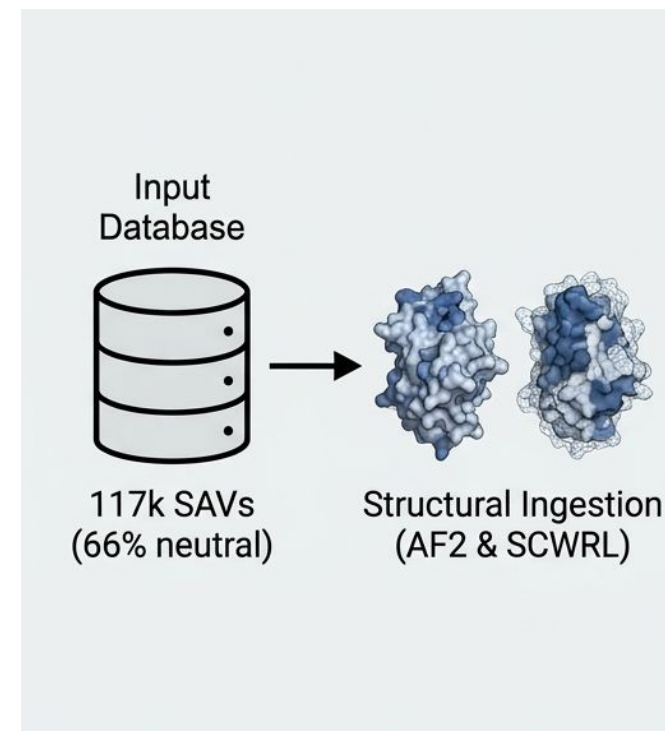
- ❖ **Single amino acid variants (SAVs)** significantly impact protein function, causing genetic disorders, cancer, and neurological conditions.
- ❖ Predicting pathogenicity from sequence alone, which was done extensively before AlphaFold2, is not enough to achieve SOTA results, which is why Rhapsody-2 incorporated a framework that uses sequential, structural and dynamical features to predict SAV pathogenicity.
- ❖ Rhapsody-2 uses XGBoost (Extreme Gradient Boosting), a decision tree based method which is easily interpretable, to predict pathogenicity based on 100 features.



Dataset Curation

ClinVar + AlphaFold2

- ❖ The Rhapsody-2 training database is built using the **ClinVar database** (December 2023), encompassing 117,525 single amino acid variants (SAVs) across 12,094 human proteins.
- ❖ **Variant labeling:**
 - "Benign" or "Likely benign" as negative (neutral, 66%),
 - "Pathogenic" or "Likely pathogenic" as positive (pathogenic, 34%),
 - discarding entries with conflicting classifications.
- ❖ Every variant is mapped to its corresponding high-accuracy wild type (WT) protein structure retrieved directly from the AlphaFold database.
- ❖ To physically place the mutated residue onto the 3D backbone, they optimized side-chain rotameric states for both the WT and mutant structures using Side-Chain Rotamer Optimization (SCWRL).



Dataset Splitting

Protein Stratification

- ❖ Evaluated using 10-fold protein-stratified cross-validation to strictly prevent data leakage and performance overestimation caused by variants of the same protein appearing simultaneously in training and test sets.
- ❖ Comparative analyses demonstrate that traditional simple cross-validation protocols artificially inflate AUROC and MCC performance metrics.

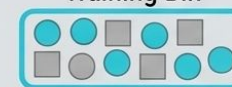
Preventing Data Leakage: The Testing Methodology

XGBoost (eXtreme Gradient Boosting) with strict 10-fold protein-stratification.

Standard Simple CV

Protein A Variants

Training Bin



Testing Bin



Data Leakage / Artificially Inflated AUROC

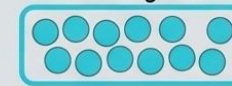
Variants of the same protein appear in both sets.

Protein-Stratified CV

Protein A Variants

Protein B Variants

Training Bin



Testing Bin



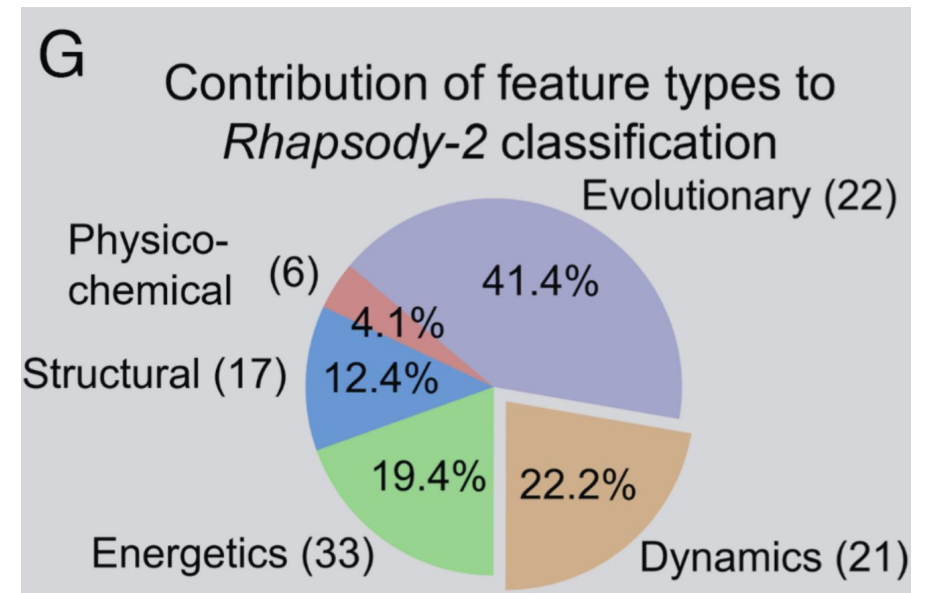
True Generalization

Guarantees accurate real-world performance metrics.

Features

Overview

- ❖ A comprehensive set of 100 computational descriptors categorized into five classes to capture structural, dynamic, and sequence attributes:
 - Sequence evolutionary features: 22 descriptors,
 - Intrinsic dynamics: 21 descriptors,
 - Folding energetics: 33 descriptors,
 - Local structural interactions: 17 descriptors,
 - Residue-specific physicochemical attributes: 6 descriptors,
 - (plus 1 intermolecular interaction feature).

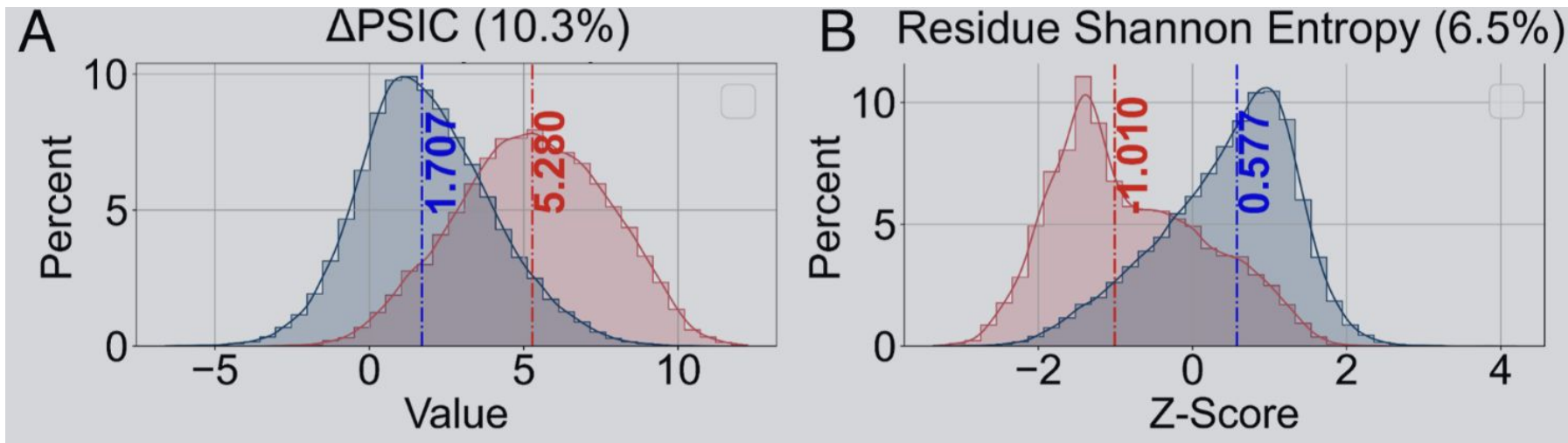


Calculated by tree split importance scores.

Features

Sequence Evolutionary Features

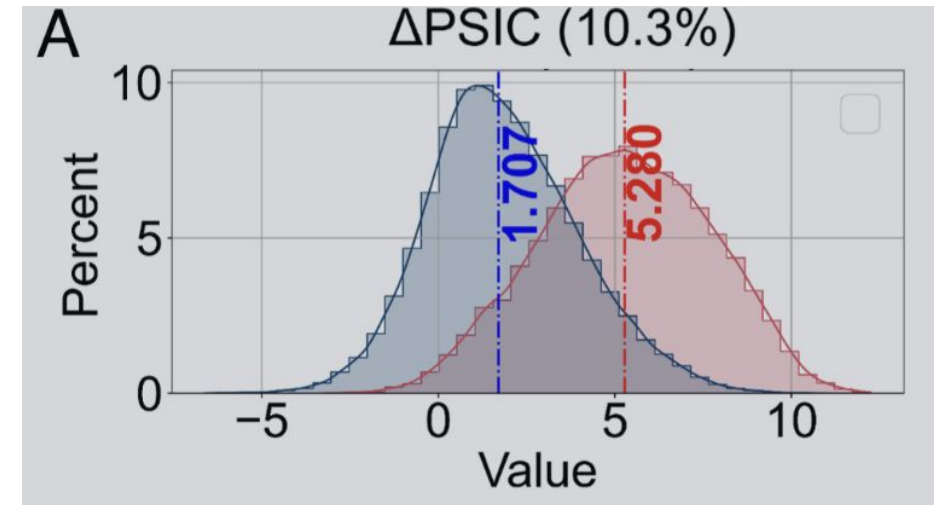
- ❖ Sequence evolutionary features consist of 22 descriptors, with a total feature contribution at 41.4% during protein-stratified cross-validation.
- ❖ They are derived via DIAMOND package and BLAST searches across the NCBI non-redundant database with a minimum 30% sequence coverage threshold.
- ❖ Key features:
 - Position-Specific Independent Counts (PSIC),
 - Shannon Entropy z-scores (with and without gaps) evaluated across sequence homologs,
 - BLOSUM62 substitution matrix scores.



Features

Sequence Evolutionary Features: PSIC

PSIC (Position-Specific Independent Counts) calculation pipeline is designed to quantify **evolutionary conservation and tolerance for amino acid substitutions**.



1. **Generating Multiple Sequence Alignments (MSAs)**
 - a. Homology Search: BLAST across the NCBI database for all target human proteins using the DIAMOND.
 - b. Filtering: Only homologs above 30% sequence coverage threshold are retained to construct the MSA.
2. **Computing Profile Matrices & Independent Counts**
 - a. Sequence Weighting: The PSIC algorithm calculates "independent counts" at each position in the alignment by weighting sequences based on their evolutionary tree distance.
 - b. Gapped Alignments: The computations explicitly allow gapped MSAs.
3. **PSIC Derivations: 9 distinct PSIC-based descriptors**
 - a. Residue-Level Scores: Wildtype, mutant and delta (WT-mutant) PSIC scores for a specific amino acid substitution.
 - b. Physicochemical Group Scores: Evaluates PSIC metrics after grouping amino acids by chemical classification (hydrophobic, polar, acidic, basic, or aromatic).
 - c. Size Group Scores: Evaluates PSIC metrics after grouping amino acids by side-chain molecular size.

Features

Sequence Evolutionary Features: Shannon Entropy & BLOSUM Score

Shannon Entropy is computed for each residue position in the alignment to **quantify local sequence conservation versus variability**.

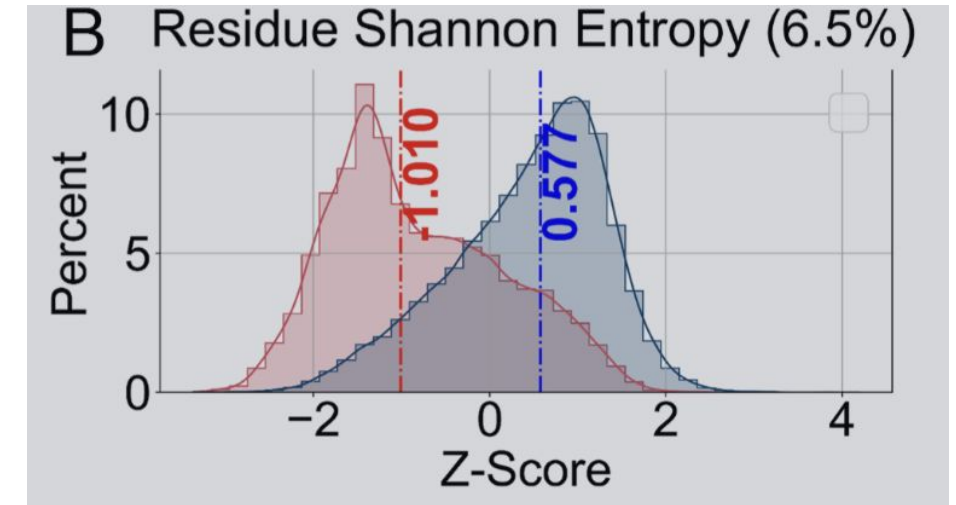
1. Computing Sequence Variability & Standardization

- Alignment Treatment: Computations are explicitly calculated across both gapped and ungapped MSAs to evaluate conservation under both conditions.
- Z-Score Normalization: Normalized Z-scores are calculated for wild-type (WT) residues to evaluate how typical or atypical a given residue's variation is relative to its sequence context.

2. Shannon Entropy Derivations: 12 Distinct Descriptors

- Residue-Level Scores: Wild-type (WT) Shannon Entropy values and corresponding Z-scores evaluated across both gapped and ungapped alignments.
- Physicochemical Group Scores: Shannon Entropy and Z-scores computed after grouping amino acids by chemical classification (hydrophobic, polar, acidic, basic, or aromatic) for gapped and ungapped MSAs.
- Size Group Scores: Shannon Entropy and Z-scores computed after grouping amino acids by side-chain molecular size for gapped and ungapped MSAs.

The BLOSUM score is derived directly from the standard **BLOSUM62** substitution matrix using the specific substitution pair: **BLOSUM(WT, Mut)**.



Features

Sequence Evolutionary Features

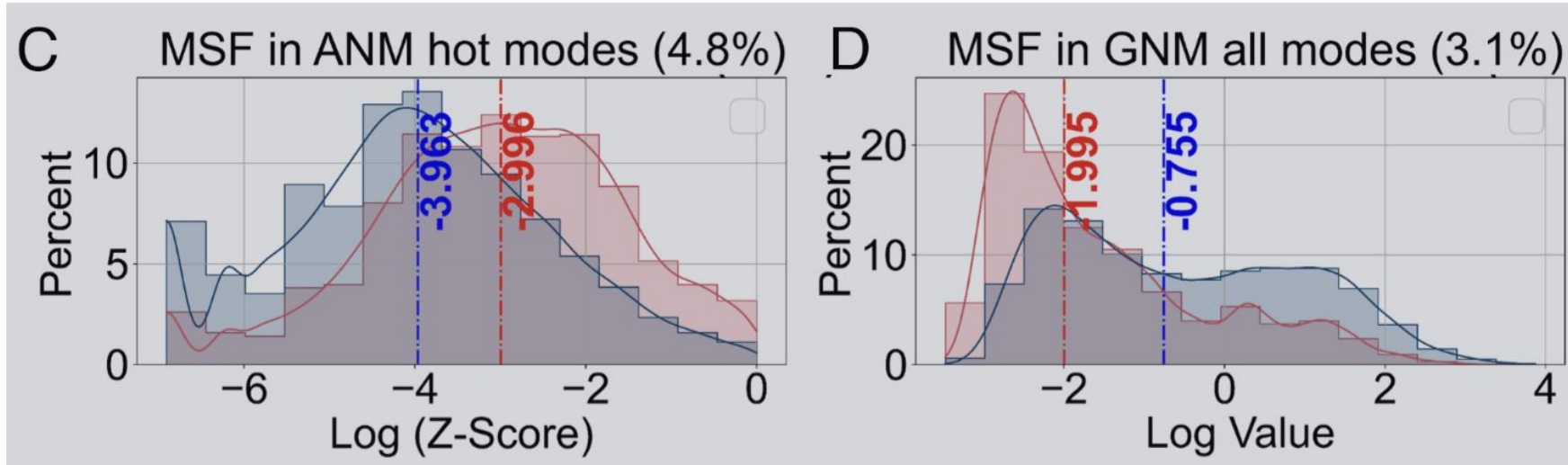
Feature #	Feature Name	10-fold Protein-Stratified Contribution (%)
1	WT-Mut Seq_PSIC (with gaps)	9.31
2	WT zscore Seq_Shannon Entropy (without gaps)	6.15
3	WT Size Group zcore Seq_Shannon Entropy (without gaps)	6.06
4	Mut Seq_PSIC (with gaps)	2.74
5	WT Size Group Seq_Shannon Entropy (without gaps)	3.36
6	BLOSUM (WT, Mut)	2.01
7	WT Size Group Seq_PSIC (with gaps)	1.89
8	WT PhyChem Group zcore Seq_Shannon Entropy (without gaps)	1.70
9	WT Seq_PSIC (with gaps)	1.17
10	WT PhyChem Group Seq_Shannon Entropy (without gaps)	0.56
11	WT Seq_Shannon Entropy (without gaps)	0.93

12	WT-Mut Size Group Seq_PSIC (with gaps)	0.79
13	WT-Mut PhyChem Group Seq_PSIC (with gaps)	0.76
14	WT PhyChem Group Seq_PSIC (with gaps)	0.75
15	WT zscore Seq_Shannon Entropy (with gaps)	0.42
16	WT Seq_Shannon Entropy (with gaps)	0.44
17	WT Size Group Seq_Shannon Entropy (with gaps)	0.43
18	WT PhyChem Group Seq_Shannon Entropy (with gaps)	0.41
19	WT PhyChem Group zcore Seq_Shannon Entropy (with gaps)	0.39
20	WT Size Group zcore Seq_Shannon Entropy (with gaps)	0.41
21	Mut PhyChem Group Seq_PSIC (with gaps)	0.37
22	Mut Size Group Seq_PSIC (with gaps)	0.36

Features

Dynamics Features

- ❖ Intrinsic dynamics features consist of 21 descriptors that capture equilibrium collective motions, structural flexibility, and allosteric communication properties of the wild-type protein structure.
- ❖ These are derived using Elastic Network Models (ENMs) implemented via the ProDy API, specifically Gaussian Network Model (GNM) and Anisotropic Network Model (ANM).
- ❖ Key features are calculated across all modes, softest modes, top 5 modes, and high-frequency ANM hot modes, they include:
 - Mean-square fluctuations (MSFs),
 - Perturbation response scanning (PRS) metrics.



Blue: neutral
Red: deleterious

Features

Dynamics Features: Model Setup

❖ **Generating Elastic Network Models (ENMs)**

- **Structural Input:** Native, wild-type (WT) 3D protein structures retrieved from AlphaFold database.
- **Models:** Structural dynamics are computed using the **Gaussian Network Model (GNM)** and **Anisotropic Network Model (ANM)** via the **ProDy API**.
- **Mathematical Framework:** The inter-residue contact topology constructs a **Kirchhoff matrix (GNM)** or **Hessian matrix (ANM)** to determine harmonic **modes of motion**.

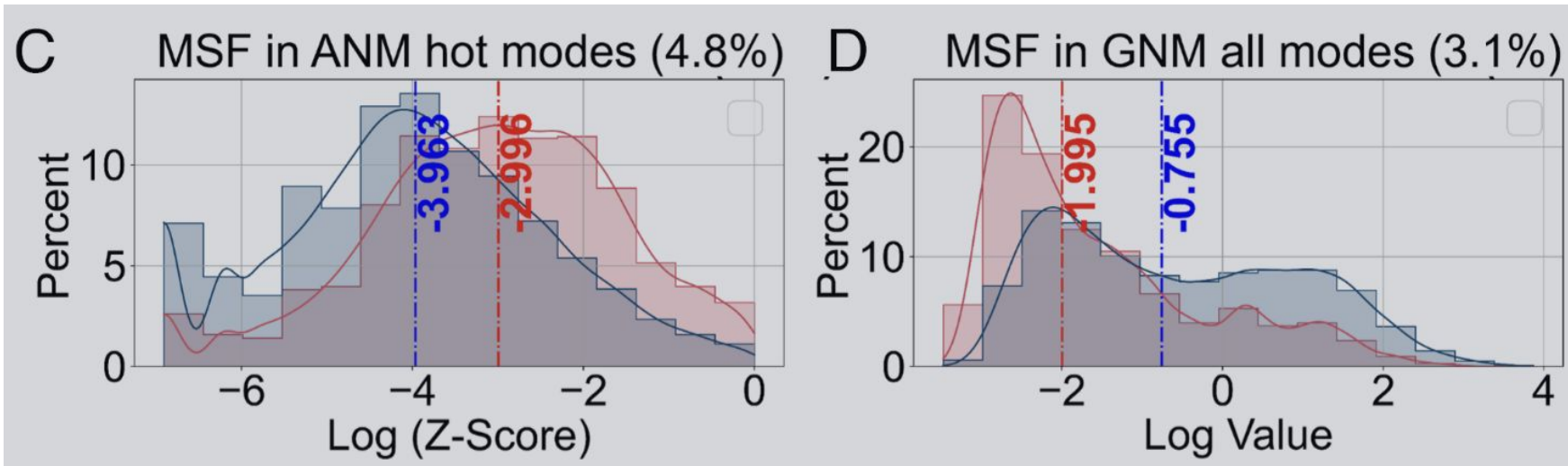
❖ **Evaluating Mode Subsets**

- **All Modes:** Captures baseline thermal equilibrium flexibility across the entire structure.
- **High-Frequency (Hot) Modes (Top 2%):** Identifies tightly packed **core residues** (kinetically hot sites) that act as folding nuclei and maintain structural stability.
- **Low-Frequency (Soft) Modes (Softest / Top 5):** Captures large-scale, highly cooperative whole-structure motions and functional rearrangements. (**allosteric sites**)

Features

Dynamics Features: MSF

- ❖ Mean Square Fluctuations (MSF) Derivations:
 - Residue-Level Descriptors: 13 individual MSF-based features such as res_MSF_zscore (ANM hot), res_MSF (GNM all), res_MSF (ANM hot), res_MSF (GNM soft), and their normalized Z-scores.
- ❖ Biological Significance:
 - Highly constrained sites (low MSF in all modes) or kinetically hot core sites (high MSF in hot modes) strongly correlate with pathogenicity when mutated due to structural destabilization.



Features

Dynamics Features: Effectiveness

❖ Perturbation Response Scanning (PRS)

- Simulated Perturbation: Directed external forces (perturbations) are applied sequentially to individual residue positions to simulate structural stress or mutational energy impacts.
- Measuring Signal Transmission (Effectiveness): **Effectiveness** quantifies the **ability of a specific mutated site to effectively transmit allosteric signals** to the rest of the protein structure upon perturbation.

❖ Effectiveness Derivations:

- Residue-Level Descriptors: 6 effectiveness features including res_effectiveness (GNM soft), res_norm_zscore_effectiveness (GNM all), res_effectiveness (GNM all), res_zscore_effectiveness (GNM all), and res_effectiveness (ANM all).

❖ Biological Significance:

- Pathogenic variants exhibit higher median effectiveness values than neutral variants (e.g., median of 0.71 vs. 0.27 for GNM soft), proving that residues acting as key allosteric signal effectors are highly susceptible to disease-causing disruptions.

Features

Dynamics Features

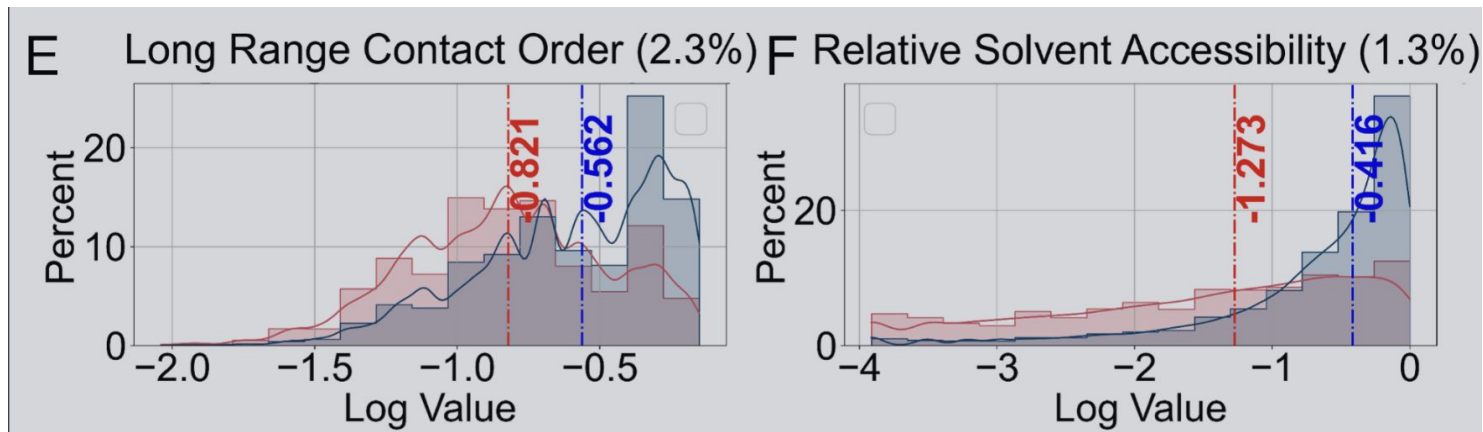
Feature #	Feature Name	10-fold Protein-Stratified Contribution (%)
23	res_MSF_zscore (ANM hot)	5.14
24	res_MSF (GNM all)	2.76
25	res_MSF (ANM hot)	2.48
26	res_MSF (GNM soft)	1.72
27	res_MSF_norm_zscore (GNM soft)	0.95
28	res_MSF_norm_zscore (ANM hot)	1.03
29	res_MSF_norm_zscore (GNM all)	0.90
30	res_MSF (GNM top5)	0.79
31	res_effectiveness (GNM soft)	0.65
32	res_norm_zscore_effectiveness (GNM all)	0.59
33	res_MSF_norm_zscore (GNM hot)	0.60

34	res_avg_meancorr (GNM soft)	0.54
35	res_MSF (ANM all)	0.54
36	res_MSF (ANM soft)	0.50
37	res_zscore_effectiveness (GNM all)	0.50
38	res_effectiveness (GNM all)	0.51
39	res_effectiveness (ANM all)	0.43
40	res_norm_zscore_sensitivity (ANM all)	0.43
41	res_MSF_zscore (GNM hot)	0.41
42	res_zscore_sensitivity (ANM top5)	0.41
43	res_hinge_binary (GNM soft)	0.32

Features

Energetics, Structure, and Local Interactions

- ❖ Folding energetics features consist of 33 descriptors calculated using the empirical **FoldX force field** (ΔG of folding and 15 individual energy components for wild-type and mutant) alongside predicted $\Delta\Delta G$ from **PROTSPOM**.
- ❖ Structural and local interaction features comprise 17 descriptors capturing spatial packing and environment after **optimizing side-chain rotameric states using SCWRL**.
 - Weighted Contact Number (WCN),
 - Weighted Hydrophobic Contact Number (WHbCN),
 - Aromatic Cluster Score (ACS), Hydrophobic Core Score (HBCS),
 - Long-Range Contact Order (LRCO),
 - Relative Solvent Accessibility (RSA),
 - Counts for hydrogen bonds, salt bridges, and disulfide linkages.



Blue: neutral
Red: deleterious

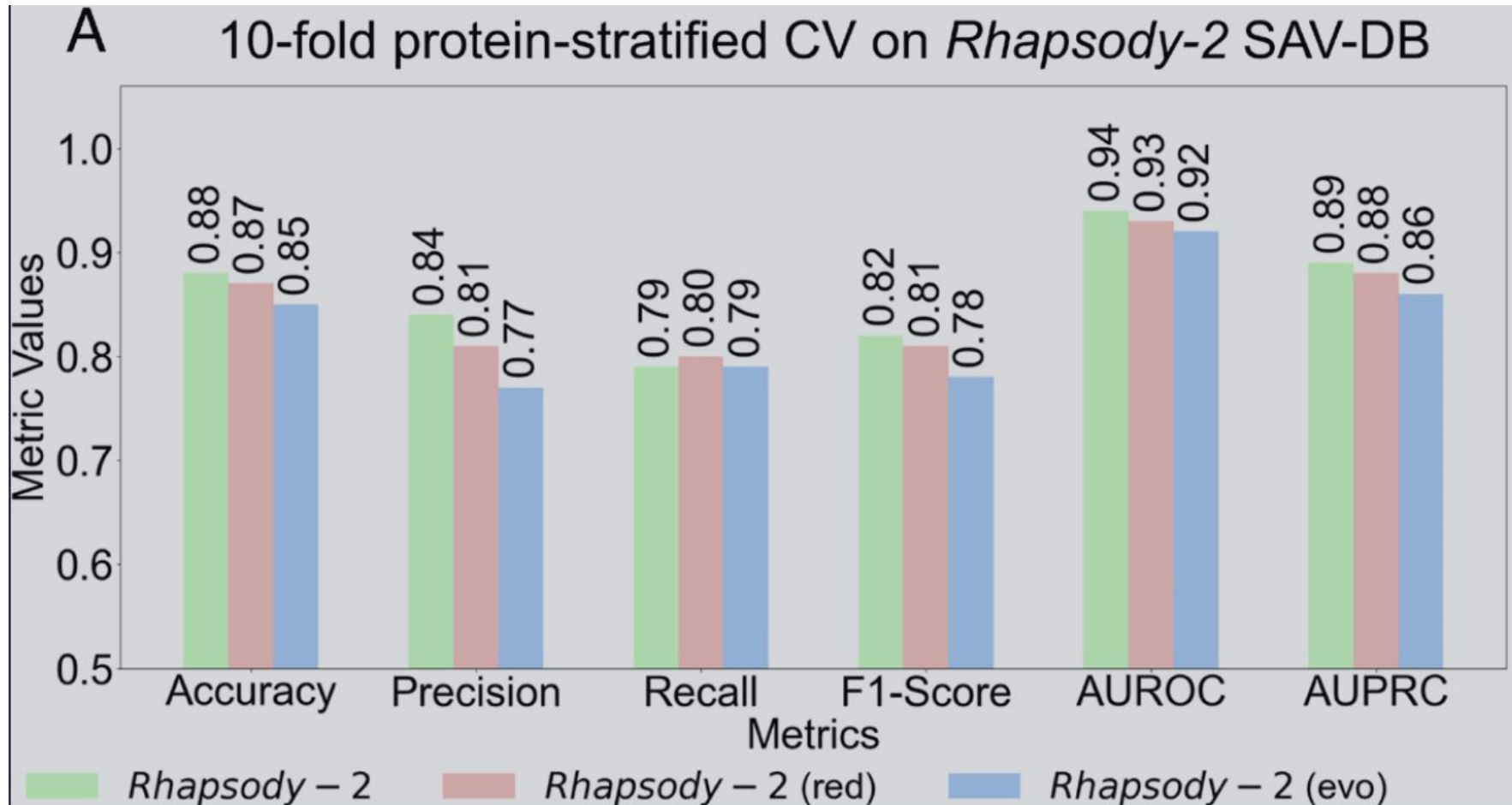
Results

Rhapsody-2 Variants

- ❖ Variants:
 - Rhapsody-2 (full model): evolutionary, dynamic, structural, energetics, physicochemical, intermolecular
 - Rhapsody-2 (red): 22 evolutionary, 21 intrinsic dynamics, 1 Relative Solvent Accessibility
 - Rhapsody-2 (evo): 22 evolutionary
 - Rhapsody-2 (dyn): 21 intrinsic dynamics
 - Rhapsody-2 (struct): 17 local structural interaction

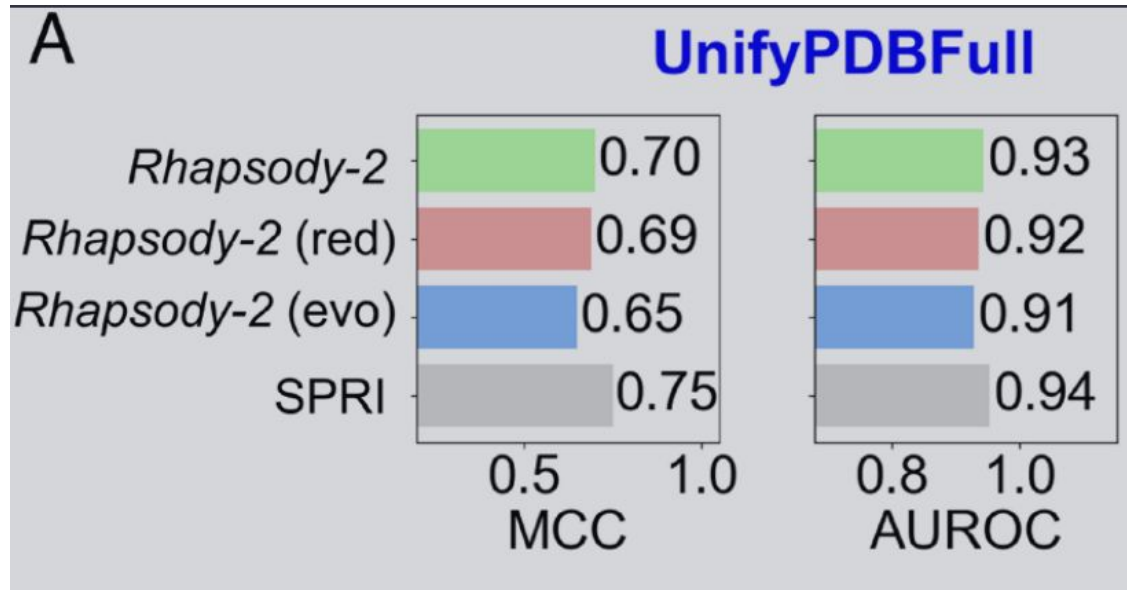
Results

On Rhapsody-2 CV



Results

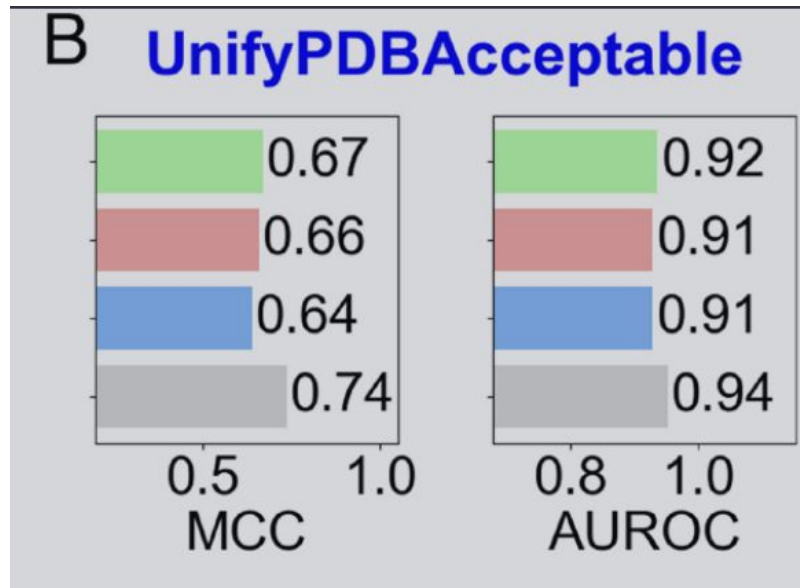
On UnifyPDB



Dataset	Method / Model Variant	MCC	AUROC
UnifyPDBFull	PROVEAN	0.65	0.89
	Polyphen-2	0.69	0.92
	PMUT	0.43	0.77
	LIST	0.49	0.83
	FATHMM	0.29	0.69
	EVmutation	0.69	0.78
	SPRI	0.75	0.94
	Rhapsody	0.73	0.92
	Rhapsody-2	0.78	0.96
	Rhapsody-2 (red)	0.74	0.94
	Rhapsody-2 (evo)	0.71	0.92
	Rhapsody-2 (dyn)	0.59	0.88

Results

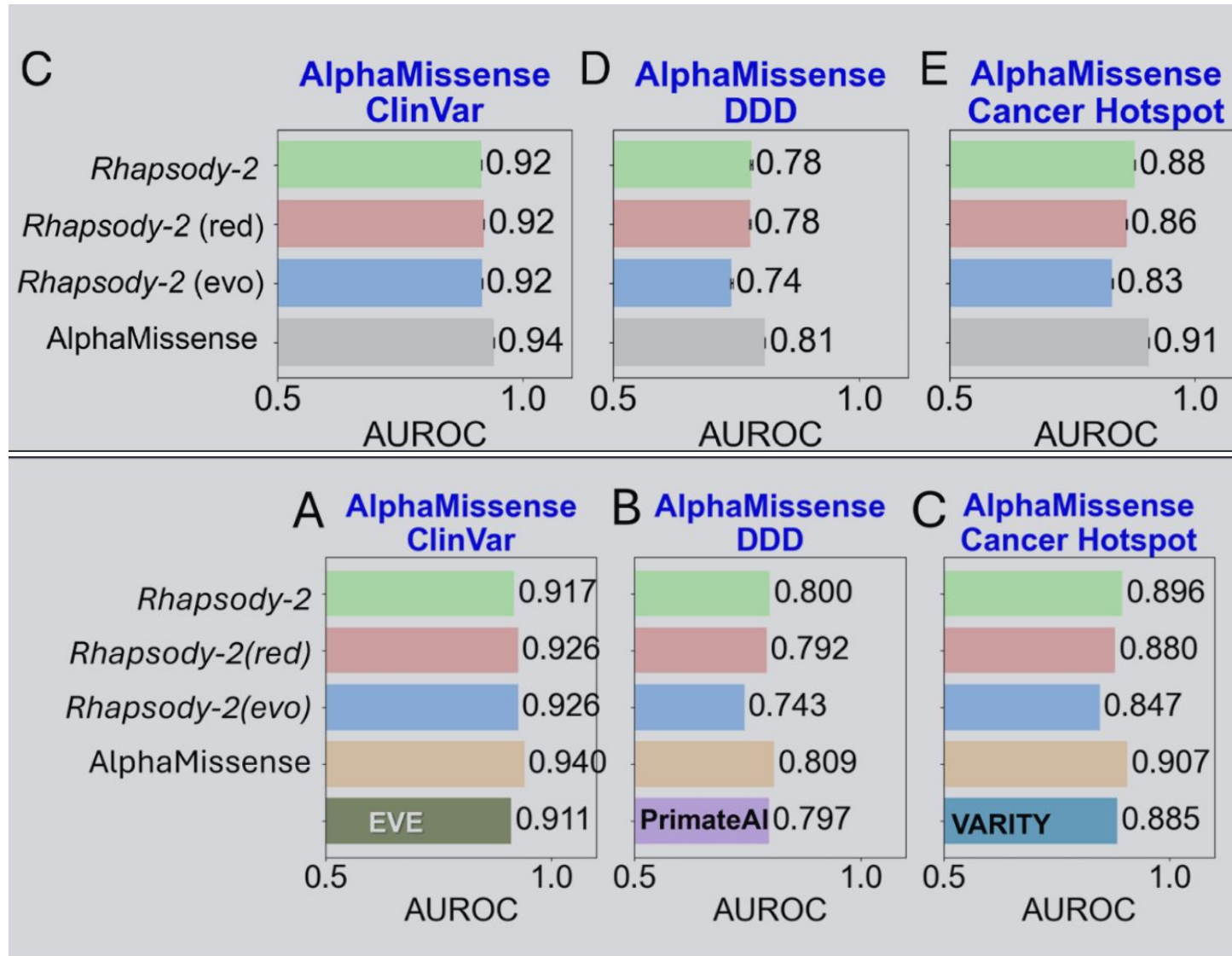
On UnifyPDB



Dataset	Method / Model Variant	MCC	AUROC
UnifyPDBAcceptable	PROVEAN	0.63	0.89
	Polyphen-2	0.69	0.92
	PMUT	0.47	0.80
	LIST	0.51	0.84
	FATHMM	0.28	0.67
	EVmutation	0.68	0.75
	SPRI	0.74	0.94
	Rhapsody	0.72	0.92
	Rhapsody-2	0.75	0.95
	Rhapsody-2 (red)	0.72	0.93
	Rhapsody-2 (evo)	0.69	0.92
	Rhapsody-2 (dyn)	0.55	0.86

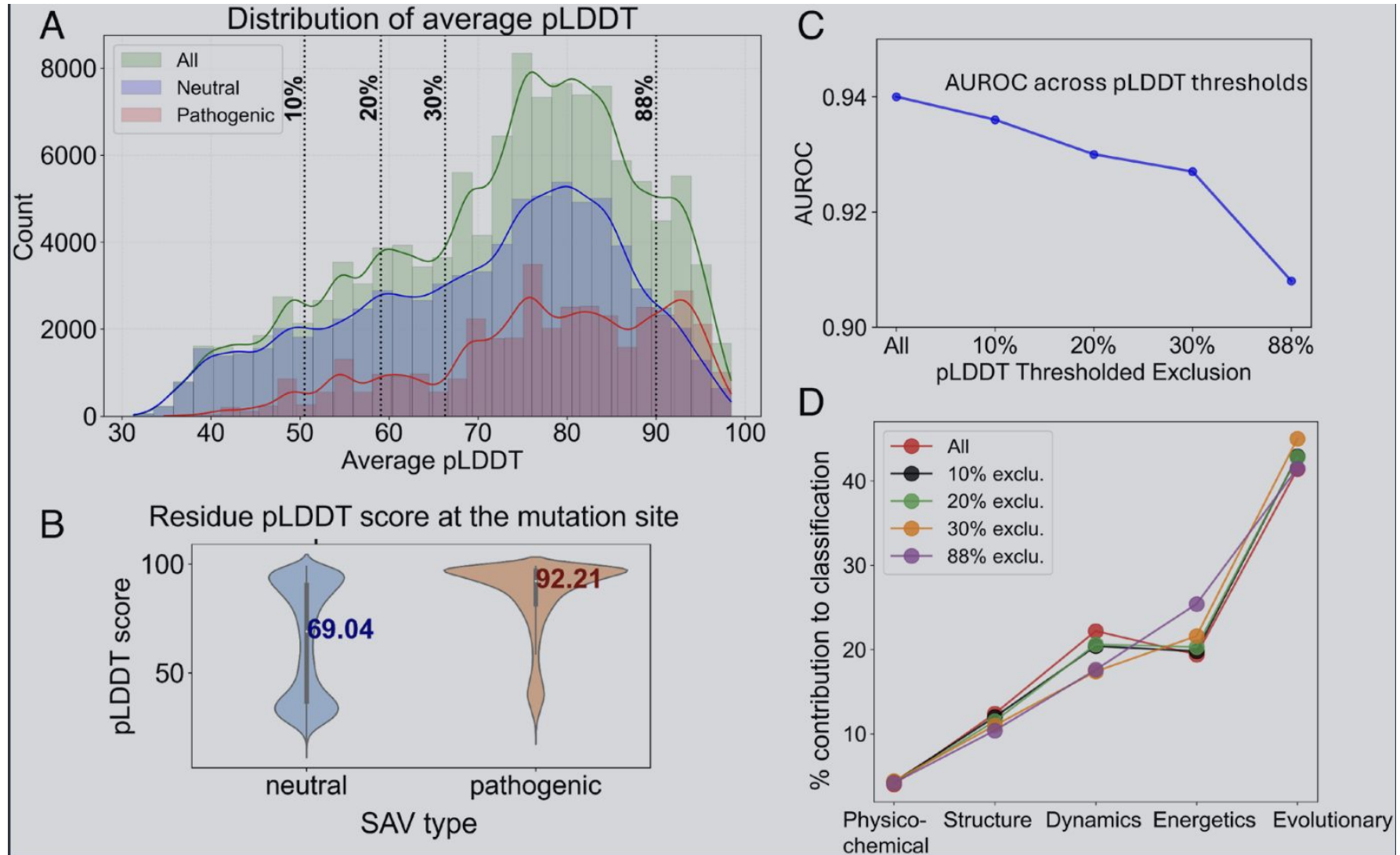
Results

Protein-Stratified vs Non stratified Results



Results

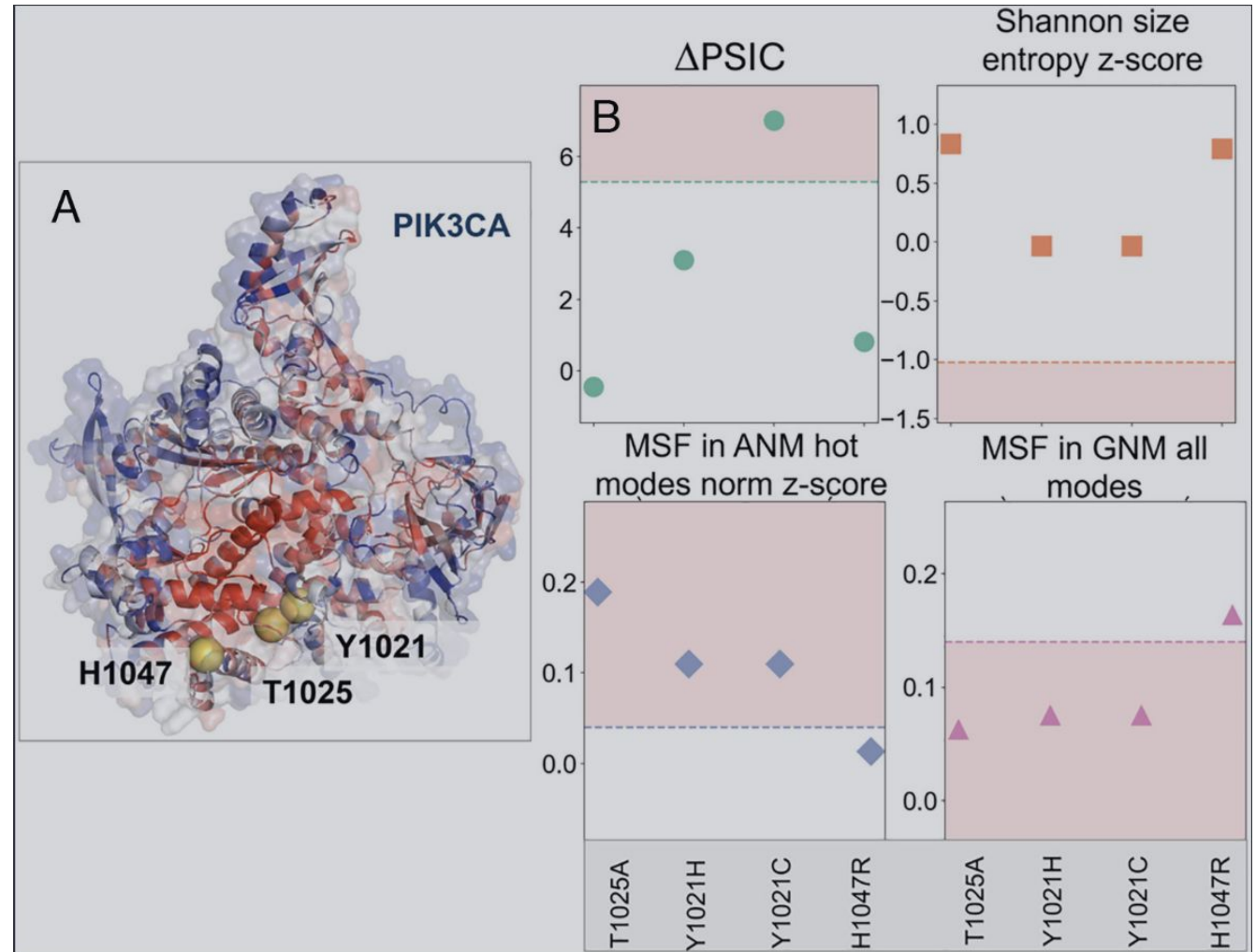
pLDDT vs Pathogenicity



Case Study (PIK3CA)

Contribution of Dynamic Features

- ❖ Applied to the **PIK3CA kinase domain**, Rhapsody-2 successfully predicts the **pathogenicity of clinical variants T1025A, Y1021H, Y1021C, and H1047R**. Analysis reveals that restricted MSFs in ANM hot modes and GNM all-modes accurately elucidate the molecular origin of pathogenicity for most of these mutations.



Thank you for listening...